Semantic Systematicity: A New Perspective on Compressibility and Expressivity in Iterated Learning

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A central question in the study of language evolution concerns how languages come to look the way they do. Language learning has been shown to be a key mechanism underlying the emergence of linguistic structure, but there are competing accounts about the cognitive biases that act on language throughout the learning process. Existing work has focused on biases towards compressibility and expressivity, which together lead to highly-structured categories. We present a novel account in which learning leads to a bias for **semantic systematicity**: a pressure for highly-structured **category systems**. We propose a formalization of this bias as an information-theoretically grounded trade-off between pressures for compressibility and expressivity that operates at a higher level than the trade-off considered in previous work. We show that our account can better explain how languages evolve in an iterated learning setting, thus shedding new light on how languages come to look the way they do.

1. Introduction

Languages exhibit a surprising amount of regularity in how they use words to carve up the world into categories. An important question is how such regularity arises (Bybee, 1988; Kirby, 1999; Levinson, 2012). One proposal is that linguistic structure is shaped by cultural evolution, particularly the repeated transmission of such structure across generations of users (Tomasello, 2010; Kirby, Griffiths, & Smith, 2014). This is supported by experimental work in the iterated learning (IL) paradigm, which simulates cultural evolution using a chain of individuals who each learn a language by observing the linguistic behaviour of the person before them (Kirby, Cornish, & Smith, 2008; Silvey, Kirby, & Smith, 2019). Over time, the language gradually adapts to the cognitive biases of its learners, leading to the emergence of **structural regularity** (Kirby, Tamariz, Cornish, & Smith, 2015; Carr et al., 2020). This is visualized in Figure 1, where initially noisy systems come to exhibit structured word-meaning mappings over time. However, there are conflicting accounts about which cognitive biases cause such regularity to emerge.

A common theoretical starting point is that languages are shaped by functional pressures for **compressibility** and **expressivity** (Kemp, Xu, & Regier, 2018): languages trade off being easy to mentally represent (maximizing compressibility)

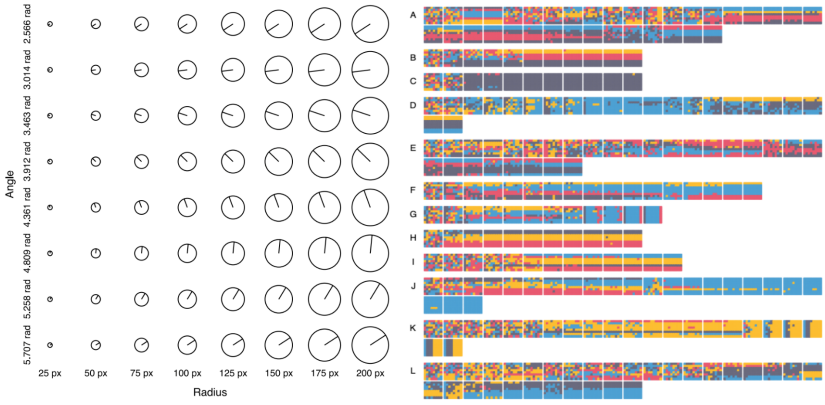

Figure 1. (a) Grid of Shepard circle stimuli (figure from Carr et al. (2020)). (b) Data for 12 IL chains. Each square is a participant's word-meaning mapping for the stimuli grid in (a), and each colour corresponds to a particular word (data from Carr et al. (2020), figure from Imel et al. (2025)).

while simultaneously enabling precise communication (maximizing expressivity). While this trade-off has been shown to constrain variation cross-linguistically (Zaslavsky, Kemp, Regier, & Tishby, 2018; Kemp et al., 2018), there are conflicting accounts of the role these biases play during IL: whether learners are primarily biased towards 1) expressive systems (Fedzechkina, Jaeger, & Newport, 2012; Carstensen, Xu, Smith, & Regier, 2015), 2) compressible systems (Carr et al., 2020; Smith & Culbertson, 2025), or 3) systems that balance both pressures simultaneously (Imel et al., 2025). These disparate findings leave the connection between IL, cognitive biases, and emergent structure unclear.

We propose that existing formalizations of these biases fail to capture a key part of how IL shapes language: previous work has largely focused on the structure present within individual categories, without capturing **system-level regularity across categories**. To preview an example we discuss later, Language 3 in Figure 2 exhibits a high degree of system-level regularity, as it consistently splits big and small circles. We call this regularity between words and higher-level semantic features **semantic systematicity**. Such systematicity is important for supporting learnability and generalizability (Ackerman & Malouf, 2013; Hallam, Jordan, Kirby, & Smith, 2025), but has yet to be accounted for when explaining the cognitive biases underlying IL.

We develop a novel information-theoretic account of semantic systematicity, and investigate whether learners are biased towards such languages during IL. We first motivate our objective function by comparing it to the Information Bottleneck (IB) framework (Zaslavsky et al., 2018), a popular formalization of the compressibility/expressivity trade-off that has been applied to the IL case study by Imel et al. (2025). We then show that our account explains patterns in experimental

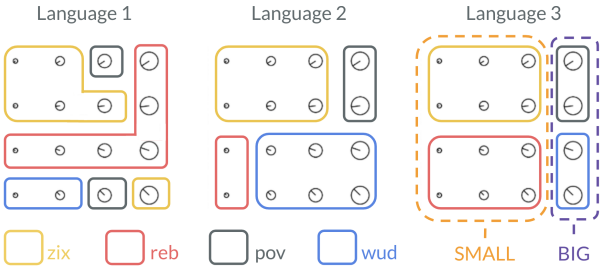

Figure 2. Three illustrative word-meaning mappings for a 4x4 grid of Shepard circles, using 4 of the nonce words from Carr et al. (2020). SMALL and BIG represent higher-level semantic groupings.

data better than the IB framework, suggesting that a bias towards semantic systematicity is at play during IL. Our findings shed new light on the debate of how cross-linguistic regularity may emerge through cultural evolutionary processes.

2. Background

Expressivity is often framed as a prediction problem: during communication, words should allow a listener to recover the speaker's intended meaning as faithfully as possible. Expressive languages often have a greater number of categories, and more expressive categories: those that group maximally similar meanings together, and so exhibit convex structure (e.g., Jäger & Van Rooij, 2007; Carr et al., 2020). All three languages in Figure 2 have the same number of categories, but Languages 2 and 3 have more expressive categories.

Formalizations of compressibility are more varied. The IB framework models the compression of a set of meanings into words (Tishby & Zaslavsky, 2015; Zaslavsky et al., 2018). Intuitively, compressible languages use fewer words, making them easier to remember. In this view, all three languages in Figure 2 are equally compressible. In contrast, the Minimum Description Length (MDL) principle (Rissanen, 1989) captures the compression of a language's intensional semantics into a set of rules (Kemp & Regier, 2012; Mollica, 2024). This notion of compressibility additionally captures category-level structure, as languages with convex categories have shorter descriptions (Fass & Feldman, 2002; Kemp & Regier, 2012). In this view, Languages 2 and 3 are more compressible than Language 1. This difference helps explain why the MDL-based method of Carr et al. (2020) suggested that IL reveals a bias for compressibility alone, while the IB-based method of Imel et al. (2025) found that both pressures were at play.

However, neither approach captures semantic systematicity: a form of system-level regularity whereby words communicate higher-level semantic features. As an example, Language 3 in Figure 2 uses words to consistently divide the stimulus grid into two higher-level features, SMALL and BIG: *zix* and *reb* indicate small circles, while *pov* and *wud* indicate big circles. This consistency plays an

important cognitive role, as the words express coarse-grained semantic distinctions that can enable a more compressed description of the semantic space. We develop a model that can capture this higher-level notion of compressibility, such that Language 3 will be the most compressible of the three languages in Figure 2.

3. Data, Baseline Model, and Our Model

Data: As in Imel et al. (2025), we use IL data collected by Carr et al. (2020) in the domain of Shepard circles. As shown in Figure 1, each stimulus is defined by values on two feature dimensions: its radius and the angle of its radial line. Each dimension has 8 discrete values, yielding a total of 64 stimuli. Carr et al. (2020) simulated 12 IL chains where each participant was shown a representative sample of half of the label-stimuli mappings produced by a previous participant, and then produced labels for the full grid. The initial systems were generated randomly using 4 nonce words, and participants were constrained to these 4 labels throughout. This process continued until convergence, for a minimum of 10 generations.

Baseline Model: The IB-based approach of Imel et al. (2025) starts with a set of meanings M that are mental representations of real world referents U – here, U is the set of 64 Shepard circle stimuli. The goal of the IB framework is to **compress** the meanings M into words W such that the words W are maximally **expressive** of the referents U. Compressibility is measured as $I(M;W)$ – the mutual information between meanings and words. More compressible systems have lower $I(M;W)$, as they can be represented with fewer bits. Expressivity is measured as $I(W;U)$ – the mutual information between words and referents. More expressive systems have higher $I(W;U)$, as words are more predictive of referents. Optimally trading-off these two pressures is equivalent to minimizing:

$$J_{IB} = I(M;W) - \beta I(W;U) \tag{1}$$

Our Model: We frame the pressure for semantic systematicity as a co-clustering problem (Kemp, Tenenbaum, Griffiths, Yamada, & Ueda, 2006) that captures a higher-level notion of the compressibility/expressivity trade-off. As in the IB framework, we compress meanings into words. The novelty of our model is that we consider an **additional kind of compression**: meanings are also grouped into higher-level features that capture coarser, language-specific semantic distinctions along a particular feature dimension (e.g., the SMALL and BIG clusters along the radius dimension in Figure 2). In languages that exhibit a high degree of semantic systematicity, words are **perfectly expressive** of these higher-level features, as the words are mapped consistently to such features. We formalize this intuition using the Generalized Information Bottleneck (GIB) framework (Slonim, Friedman, & Tishby, 2006), an extension of the IB framework that models more complex compression tasks. Here, we present high-level details of this formalization; for more details, see our Supplementary Materials, code, and data (link here: `https://github.com/jaikaggarwal/evolang_2026`).

In addition to compressing meanings M into words W, we propose to compress M into a set of higher-level features Z_D on each semantic feature dimension D. A compressible set of higher-level features requires a speaker to remember fewer semantic distinctions along a feature dimension. For example, in Figure 2, Language 3 is consistent with $Z_D = \{\text{SMALL, BIG}\}$, which is more compressed than considering all 4 radius values individually. This compression is measured as the mutual information term $I(M; Z_D)$. Our formulation of expressivity also operates on these higher-level features: in expressive systems, the words W should be maximally informative about the semantic distinctions Z_D present along each dimension D. For example, the words in Language 3 perfectly align with Z_D, but in Language 2, the word *wud* is ambiguous about these higher-level features. Mathematically, this corresponds to maximizing the mutual information term $I(W; Z_D)$. We can repeat this compressibility/expressivity trade-off for each feature dimension of interest. Shepard circles have two such features: the angle A of a radial line and radius R. Minimizing the compression-based terms and maximizing the expressivity-based terms leads to the following objective function:

$$J_{GIB} = I(M; W) + I(M; Z_A) + I(M; Z_R) - \beta[I(W; Z_A) + I(W; Z_R)] \quad (2)$$

where the parameter β modulates the trade-off. Languages that are semantically systematic minimize J_{GIB}, while languages with irregular structure deviate from optimality. Following Zaslavsky et al. (2018), we measure a language L's deviation from optimality ϵ as the normalized difference between its objective function value J^L_{GIB} to the theoretically optimal value J^*_{GIB} at a particular value of β: $\epsilon = \min_\beta \frac{1}{\beta}[J^L_{GIB} - J^*_{GIB}]$. This quantity ϵ, computed analogously for J_{IB}, allows us to test key predictions about our model.

4. Analysis & Results

We hypothesize that semantic systematicity, formalized in our novel trade-off between compressibility and expressivity of higher-level semantic features, will explain the IL data better than the standard IB formulation used by Imel et al. (2025). We test this hypothesis by considering the deviation from optimality of each model (ϵ explained above), used as a measure of model fit in the IB literature (Zaslavsky, Kemp, Tishby, & Regier, 2020).

The plot in Figure 3 shows how the deviation from optimality ϵ changes over generations for each IL chain in Figure 1. The grey and red lines indicate ϵ values under the standard IB objective J_{IB} and our novel objective J_{GIB}, respectively. First, focusing on the red lines in Figure 3, we see that under J_{GIB}, 11 out of the 12 chains converge to perfectly optimal solutions (i.e., $\epsilon = 0$). The exception is Chain C, where the final participant used a second word to label a single meaning in an otherwise optimal 1-word system. That systems almost always converge to perfectly optimal solutions under our objective function provides evidence that languages are shaped by a pressure for semantic systematicity during IL.

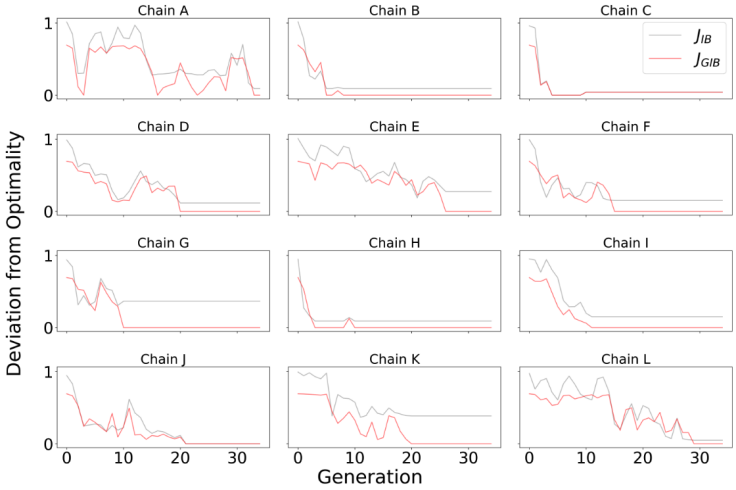

Figure 3. Deviation from optimality across generations according to our model J_{GIB} (red) and the baseline model J_{IB} (grey). Flat lines indicate convergence prior to the maximum of 34 generations.

Second, comparing the red and grey lines in Figure 3, we see that our model and the baseline model make similar predictions in earlier generations. However, once systems converge, ϵ is almost always lower under J_{GIB} than under J_{IB}. Given that linguistic systems arising through IL are considered closer to optimal under our objective function, this provides evidence that the structure that emerges through IL is better explained by a bias towards semantic systematicity.

5. Discussion

We develop a novel information-theoretic objective function that more directly quantifies the degree of system-level structural regularity present in language: its degree of **semantic systematicity**. We find that throughout the process of iterated learning, languages tend to minimize our new objective function, before converging on optimal solutions. These findings suggest that IL reveals a cognitive bias for a compressibility/expressivity trade-off, but over different cognitive constructs than words and meanings; this trade-off operates at a more abstract level, between words and higher-level semantic features. It is worth noting that we are not claiming that the standard trade-off has no role in shaping language. However, in the case of this IL task where participants did not engage in any substantive communicative task, it is more likely that semantic systematicity (which has been shown to support learnability and generalizability) plays a more important role than the standard trade-off (which has been taken to support communication). In future work, we aim to test our theory on more naturalistic lexical semantic systems, to help shed light on how the two levels of the trade-off jointly shape language.

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